europass

Naveen Krishnan Radhakrishna Pillai

Passport: Y8517752 | **Date of birth:** 29/09/2002 | **Phone number:** (+91) 9207176317 (Mobile) |

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EDUCATION AND TRAINING

10/10/2021 – 09/08/2025 Amritapuri, India

BACHELOR OF TECHNOLOGY IN ELECTRONICS AND COMPUTER ENGINEERING Amrita Vishwa Vidyapeetham, Amritapuri, Kerala

Website https://www.amrita.edu/campus/amritapuri/ | **Final grade** CGPA : 7.71 | **Number of credits** 162 |

Thesis Microwave imaging for Tumor Detection using Antennas

2019 – 2021 Changanassery, India

HIGHER SECONDARY Good Shepherd Public School

Final Grade - 88%

Website https://goodshepherdjc.com/

2015 – 2019 Adoor, India

SECONDARY Kendriya Vidyalaya Sangathan

Final Grade - 85.8%

Website https://kvsangathan.nic.in/en/kv-school/kendriya-vidyalaya-adoor/

WORK EXPERIENCE

INTERN – CENTRE FOR FLEXIBLE ELECTRONICS AND ADVANCED MATERIALS – 06/01/2025 – 06/07/2025 – AMRITAPURI

Tumor Monitoring and Response System, a web-based platform for real-time tumor detection and statistical reports for doctors

INTERN – AMMACHI LABS – 01/08/2024 – 01/02/2025 – AMRITAPURI, INDIA

Conducted coconut disease research using drone-based data collection combined with systematic field surveys. Focused on gathering high-quality data to support accurate analysis and decision-making.

INTERN – TRANQUILITY IOT AND BIG DATA SOLUTIONS (TIBS) – 01/09/2023 – 01/08/2024 – AMRITAPURI, INDIA

Tested the range of the sector and omnidirectional antenna and did coding for the sensor. Incorporating sensors to transmit data over a long distance and successfully communicating data over a 3km range.

PROJECTS

2025

heypage

Create your personal link page to neatly organize all your important links by category. Improve your digital presence, showcase portfolios and projects, highlight social profiles, and share everything in one professional place.

Links https://www.heypage.online | https://www.heypage.online/hasselx

2024 – 2025

Microwave Imaging for Tumor Detection using Antennas

Designing a System for frequent tumour monitoring and visualisation using advanced reconstruction techniques, enabling real-time, non-invasive diagnostics featuring a doctor and patient web portal with patient data stored inside a cloud for progression analysis.

Link https://tumors.onrender.com/

2023

Digital Score Updater, Microprocessor

Developed a logic system to track and update scores for multiple teams using an ARM microcontroller. Simulated the project within the Proteus environment, ensuring effective functionality for real-time score updates.

PUBLICATIONS

2026

Tumor Detection and Characterization Using Microwave Imaging Technique—An Experimental Calibration Approach

This study presents a non-invasive microwave imaging system for detecting and localizing breast tumors by exploiting contrasts in electrical properties between healthy and malignant tissues. Using horn antenna measurements and a modified beamforming algorithm, high-contrast tumor phantoms of varying sizes and positions are accurately reconstructed with a 96% detection accuracy.

Authors: Anudev Jenardanan Nair; Suraksha Rajagopalan; Naveen Krishnan Radhakrishna Pillai; Massimo Donelli; Sreedevi K. Menon | **Journal Name**: Sensors | **Volume, Issue and Pages**: Vol. 26, Issue 3, Article 1014 | **Publisher**: MDPI (Multidisciplinary Digital Publishing Institute)

2024

Design, Development, and Analysis of Spiral Microstrip Resonator Filter

Designed and validated a spiral resonator-based microstrip sensor with less than 3% error in resonance frequency prediction, demonstrating applications in RFID tags, filters, and wireless communication systems. Conducted simulations in ANSYS HFSS and prototyping using vector network analysis.

Authors: Lauksiga M P; Naveen Krishnan R; Pranav Vinod; Suraksha Rajagopalan; Sreedevi K. Menon | **Journal Name**: 2024 11th International Conference on Advances in Computing and Communications (ICACC) | **Volume, Issue and Pages**: pp. 1–4 | **Publisher**: IEEE

SKILLS

skills of prompt search for information on the Internet | Python Language - Basic knowledge | Completing DIY projects | Ability to work in a team and autonomously | MATLAB | R | Lt spice | Cisco Packet | Python (Spyder/ Jupyter Notebook IDE) | Wireshark

LANGUAGE SKILLS

Mother tongue(s): **MALAYALAM**

Other language(s): **ENGLISH**

HOBBIES AND INTERESTS

Music Movie 3D Modelling

VOLUNTEERING

2025 Amritapuri

ARISE - Amritapuri

An event conducted every year in Amrita Vishwa Vidyapeetham to honour the research and publication contributions of the staffs.

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2023 – 2024 Amritapuri

Student Social responsibility SSR

Click Smart Stay Away from Scams 2023 | Conducted an informative session for students on various cyber scams, raising awareness and imparting crucial knowledge to enhance their online safety.

2023 Amritapuri

AMMA'S 70th BIRTHDAY

Mata Amritanandamayi, the Chancellor of Amrita Vishwa Vidyapeetham, is also known as AMMA. On her 70th birthday, I got the opportunity to help and organise the events.